

MITRAL VALVE DISORDERS

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Mitral valve disorders are among the most common valvular diseases with an increasing prevalence in the aging population. It is estimated that over 10% of patients over 75 years old have significant mitral valve disease with most of these due to mitral regurgitation (MR).^{1,2} The decreasing incidence of rheumatic heart disease has made mitral stenosis increasingly rare in the United States, although it remains a problem in developing nations and is a significant global health issue.³

The hemodynamic abnormalities associated with mitral stenosis and regurgitation lead to interesting and complex challenges for physicians. Major decisions regarding the treatment of mitral valve disease are often based on these evaluations, and it is important for the physician to appreciate the limitations and potential sources of error when evaluating these conditions. Furthermore, understanding mitral valve hemodynamics is crucial as new transcatheter therapies are developed and applied to these conditions.

MITRAL STENOSIS

Rheumatic heart disease is the most common cause of mitral stenosis; other etiologies are rare (Box 5.1). Mitral annular calcification is becoming increasingly prevalent in our aging population; however, it rarely causes severe stenosis. Although rheumatic heart disease has decreased dramatically in the United States, it continues to affect populations with substandard medical care, including Mexican Americans, Native Americans, and immigrants from developing nations. Pathologically rheumatic mitral stenosis results from several mechanisms, including commissural fusion, cuspal fibrosis and thickening, and chordal fusion and thickening (Fig. 5.1).⁴ Other nonrheumatic pathologic conditions causing mitral stenosis are very rare.

The unobstructed, normal mitral valve orifice area measures approximately 4 cm². Symptoms become apparent when the valve area falls below 2 cm², and stenosis is classified as “severe” when the valve area measures <1.5 cm². The natural history of mitral stenosis is characterized by a long latent period lasting many years, with patients experiencing either no or minimal symptoms.⁴ Symptoms may arise insidiously over many years and lead to progressive disability from dyspnea or fatigue. Alternatively, a patient may experience the abrupt onset of symptoms from either the development of rapid atrial fibrillation (AF) or acute volume overload. Important complications that arise from the natural history of mitral stenosis include AF, cerebral and peripheral embolic events, hemoptysis, pulmonary hemorrhage, pulmonary hypertension and right-sided heart failure, endocarditis, and increased predisposition to infections.

PATHOPHYSIOLOGY OF MITRAL STENOSIS

Obstruction of the mitral valve causes a pressure gradient between the left atrium and left ventricle. The presence of this pressure gradient throughout diastole defines the hemodynamic hallmark of significant mitral valve stenosis. Very mild degrees of mitral stenosis have either an undetectable or very small diastolic gradient. With the progressive narrowing of the mitral valve, left-atrial pressure rises and the gradient becomes more pronounced. In addition to elevating left-atrial pressure, incomplete emptying of the left atrium impairs filling of the left ventricle and diminishes cardiac output.

Because the left atrium is in series with the pulmonary circulation, elevated left-atrial pressure passively elevates pressure in the pulmonary veins and arteries. Early in the course of the disease, pulmonary vascular resistance is normal with little effect on the right heart. As the condition progresses and becomes more chronic, the pulmonary artery pressures rise further. Pulmonary hypertension is initially due to reactive changes in the pulmonary arteriolar bed and is reversible. Marked pulmonary hypertension may result and reach systemic levels, obstructing blood flow through the lungs (the “second” stenosis of mitral stenosis), further decreasing cardiac output. Pulmonary vascular resistance increases substantially, causing enlargement of the right ventricle and right-sided heart failure. Although pulmonary hypertension is usually reversible following relief of mitral stenosis by either surgery or balloon valvotomy, with advanced, end-stage mitral stenosis, pulmonary hypertension may become fixed from permanent anatomic changes in the pulmonary arteries and arterioles.

The left-atrial pressure and the cardiac output are the main determinants of symptoms in patients with mitral stenosis. Elevated left-atrial pressure causes dyspnea, pulmonary edema, and hemoptysis. The low cardiac output associated with this condition causes fatigue. In addition to the mitral valve orifice area, left-atrial pressure depends on the rate of flow across the valve (i.e., cardiac output), heart rate, size and compliance of the left atrium, and volume status; heart rate and volume status are particularly important. If given enough time, the left atrium will eventually empty even in the presence of severe mitral stenosis. Thus for any given mitral valve area, bradycardia will result in lower left-atrial pressures, and tachycardia will result in higher left-atrial pressures. For this reason, the onset of rapid AF is poorly

Box 5.1 Causes of Mitral Valve Stenosis

1. Rheumatic heart disease
2. Mitral annular calcification
3. Congenital mitral stenosis
4. Lupus
5. Infective endocarditis/vegetation
6. Carcinoid
7. Rheumatoid arthritis
8. Methysergide therapy
9. Radiation-induced valve disease
10. Prosthetic mitral valve dysfunction

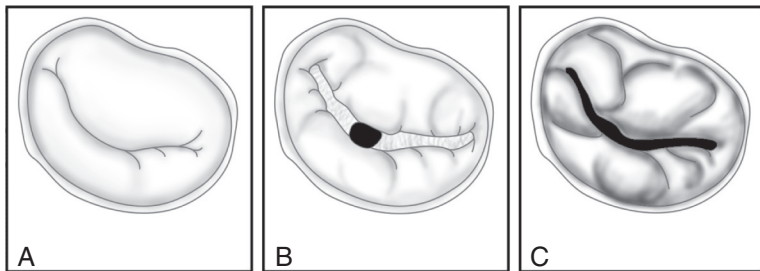


Fig. 5.1 Pathology of rheumatic mitral stenosis. (A) Normal valve. (B) A commissural fusion represents the most common pathologic mechanism of mitral stenosis, causing a “fish mouth” orifice. (C) A noncommissural form in which there is extensive fibrosis and calcification at the leaflet tips, resulting in impaired opening of the valve and stenosis.



Fig. 5.2 In patients with mitral stenosis and normal sinus rhythm, prominent *a* waves may be apparent on the left-atrial or pulmonary capillary wedge pressure tracing (arrow).

tolerated and leads to the abrupt onset of symptoms including acute pulmonary edema. Similarly acute volume overload will rapidly increase left-atrial pressure, leading to dyspnea or pulmonary edema.

HEMODYNAMICS OF MITRAL STENOSIS

A wide spectrum of hemodynamic abnormalities is possible in patients with mitral stenosis, depending on the stage of their disease. Initially the major hemodynamic abnormalities reflect solely the mitral valve obstruction and include (1) elevation of the left-atrial or wedge pressure, typically to 20–25 mm Hg, with normal or low left-ventricular

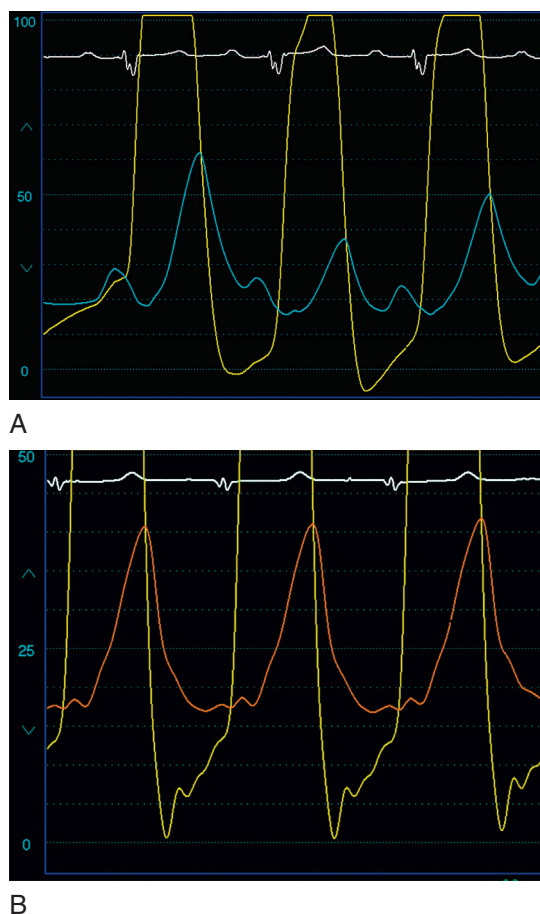


Fig. 5.3 (A and B) Examples of large *v* waves on the left-atrial pressure waveform in two patients with mitral stenosis.

end-diastolic pressure (LVEDP); (2) the presence of a pressure gradient that exists throughout diastole between the left atrium and left ventricle, usually ranging from 5 to 25 mm Hg; (3) a reduction in cardiac output (3.5–4.5 L/min); and (4) abnormalities in the left-atrial pressure tracing, affecting both *a* and *v* waves. In patients with normal sinus rhythm and mitral stenosis the *a* wave on the left-atrial or pulmonary capillary wedge pressure (PCWP) waveform may be accentuated because of the increased residual volume of the atrium at the onset of atrial systole (Fig. 5.2). The *a* wave may be quite large, and values as high as 50 mm Hg have been described.⁵ A prominent *v* wave may also be observed in pure mitral stenosis because left-atrial volume and pressure are already high, and any additional increase in volume that occurs during passive atrial filling results in a greater increase in pressure, generating prominent *v* waves (Fig. 5.3).^{5,6} There also may be a contribution to reduced left-atrial compliance from fibrosis. The presence of a large *v* wave correlates strongly with diminished exercise tolerance and is a significant predictor of pulmonary hypertension.^{7,8} Furthermore, because mitral stenosis delays emptying of the left atrium, the slope of the *y* descent, representing the phase of early and rapid ventricular filling, is delayed (Fig. 5.4) compared with the rapid descent seen in MR.

In the early stages of the disease pulmonary pressures are normal and then become modestly elevated, despite the presence of severe mitral orifice narrowing. At this stage the high pulmonary artery pressures reflect elevated left-atrial pressure; the pulmonary vascular resistance is normal. Over time, however, the pulmonary vascular resistance increases due to reactive changes, and right-ventricular enlargement occurs. Late stages of mitral stenosis are associated with marked pulmonary hypertension due to permanent anatomic changes in the arterioles, causing extreme elevations in pulmonary vascular resistance, pronounced right-ventricular failure, and severe secondary tricuspid regurgitation.

The existence of a pressure gradient between the left atrium and left ventricle during diastole both defines mitral stenosis and forms the basis of the hydraulic formula derived to calculate the mitral valve orifice area. In patients



Fig. 5.4 The *y* descent (arrow) is delayed in patients with mitral stenosis consistent with impaired emptying of the left atrium.

without mitral stenosis, left-atrial and left-ventricular diastolic pressure curves appear nearly superimposable (Fig. 5.5A). In fact a very small gradient must normally exist to allow blood to flow from the left atrium into the left ventricle (Fig. 5.5B), but this is often not appreciable by the clinically used, fluid-filled transducers. In contrast, in mitral stenosis, a pressure gradient is present immediately on opening of the mitral valve and persists in diastole so that diastasis is absent (Figs. 5.2–5.4).⁹

The transmittal gradient is ideally assessed by obtaining simultaneous pressure waveforms from catheters positioned in the left atrium and left ventricle. However, most physicians are not facile at the performance of transseptal catheterization, and the PCWP is typically substituted for left-atrial pressure. Although this practice may be acceptable in some cases, the potential for considerable error exists, and the limitations of this technique must be understood. Thus when an important decision hinges on an accurate measurement of the pressure gradient across the mitral valve, transseptal catheterization is necessary.

In general, the PCWP correlates well with left-atrial pressure.^{10,11} This is particularly true when the PCWP is low (<25 mm Hg), with no significant difference noted between left-atrial pressure and the PCWP.¹⁰ When the PCWP is >25 mm Hg, considerable error may exist (variance in excess of 10 mm Hg).

Although a good correlation exists between the mean left-atrial pressure and mean PCWP, the transmittal gradient using the PCWP does not correlate as well with the gradient obtained using the left-atrial pressure.^{12–14} Major sources of error exist; first, the PCWP introduces a considerable time delay (40–160 ms depending on the position of the catheter), and this pushes the *v* wave into diastole; and second, intrinsic to the generation of a wedge pressure, there is dampening of the PCWP waveform. In general, these factors lead to an overestimation of the pressure gradient (Fig. 5.6). In addition, in the presence of pulmonary hypertension (a common occurrence in patients with mitral stenosis), it may not be possible to obtain a true PCWP from the pulmonary artery position, and instead the waveform represents a hybrid between the true wedge pressure and the pulmonary artery pressure, and this falsely elevates the “wedge.”¹⁴ These factors conspire to elevate the mean diastolic gradient compared with that obtained with left-atrial pressure. Adjustment for the time delay by phase shifting the tracing relative to the left-ventricular pressure provides a more accurate reflection of the left-atrial–left-ventricular pressure gradient.¹³ However, several experts believe that these inaccuracies make the use of the PCWP an unreliable gauge of the transmittal gradient, and thus this method should not be used to make major decisions such as referral for mitral balloon valvuloplasty or mitral valve surgery.^{12,14} Importantly, if the PCWP is used, the operator must pay meticulous attention to detail and confirm the wedge pressure using oximetry sampling to demonstrate an arterial saturation >95%. In our practice, the use of simultaneous PCWP–left ventricular waveforms is useful to exclude the presence of a mitral gradient. However, in patients with evidence of a significant gradient using the PCWP who have discrepant noninvasive studies, poor-quality wedge pressure waveforms, pulmonary hypertension, or prior prosthetic valve surgery, transseptal catheterization should be performed to confirm the gradient with a left-atrial pressure measurement before making a major decision related to the mitral valve stenosis.

A transmittal gradient has several causes other than true mitral stenosis (Box 5.2). Severe mitral annular calcification may result in a transmittal gradient.¹⁵ This may be seen in association with calcific aortic stenosis (Fig. 5.7) or in patients with renal failure. Typically the gradient is modest (averaging 4 mm Hg by echocardiography).¹⁶ Although there is little published data, in our experience, echocardiography tends to overestimate the pressure gradient in the presence of severe mitral annular calcification. This is likely because these estimates are based on Doppler velocities and these may be increased because of annular restriction and compliance issues rather than true stenosis. In many cases,

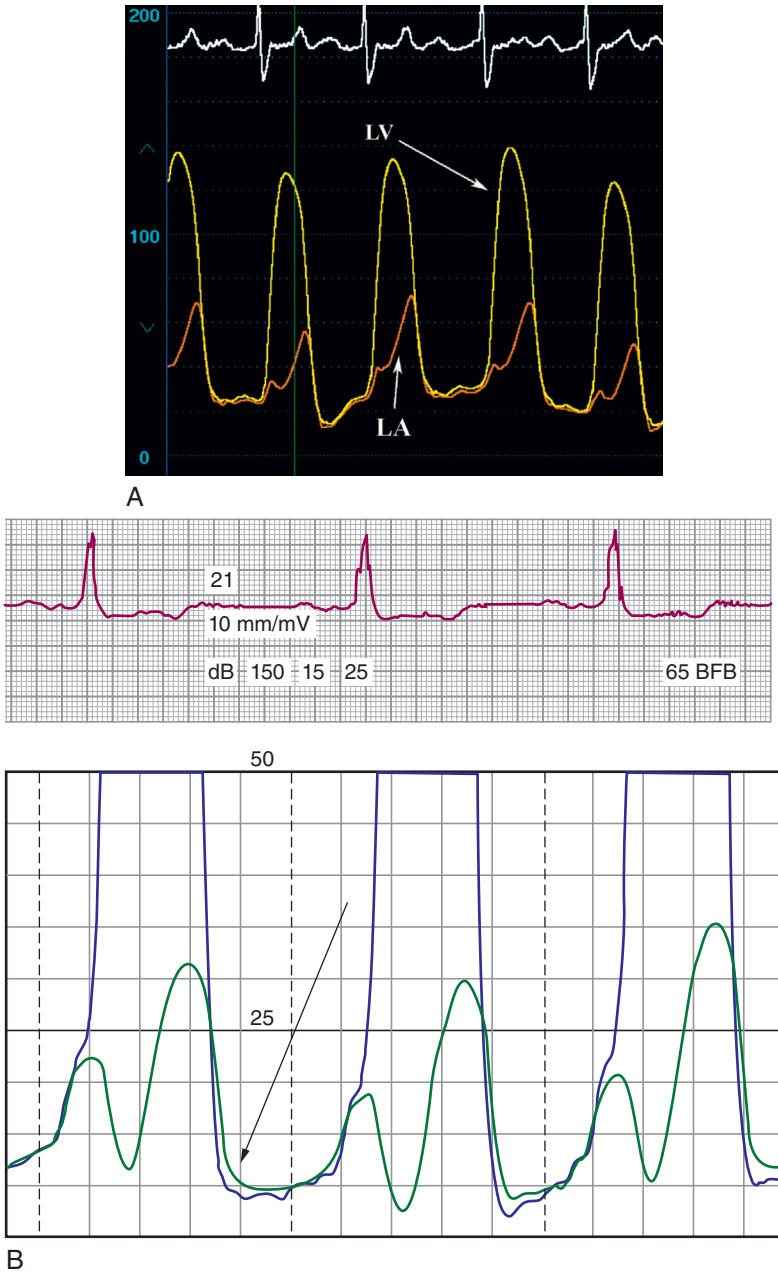


Fig. 5.5 Normal left-atrial and left-ventricular relationship. (A) A patient in atrial flutter with significant mitral regurgitation and a prominent v wave. Note that the y descent of the v wave is brisk and coincides with the downslope of the left-ventricular pressure. The diastolic pressures are virtually superimposable. (B) A small gradient between left-ventricular diastolic pressure and left-atrial pressure may be apparent early in diastole (arrow). LA, Left atrial; LV, left ventricular.

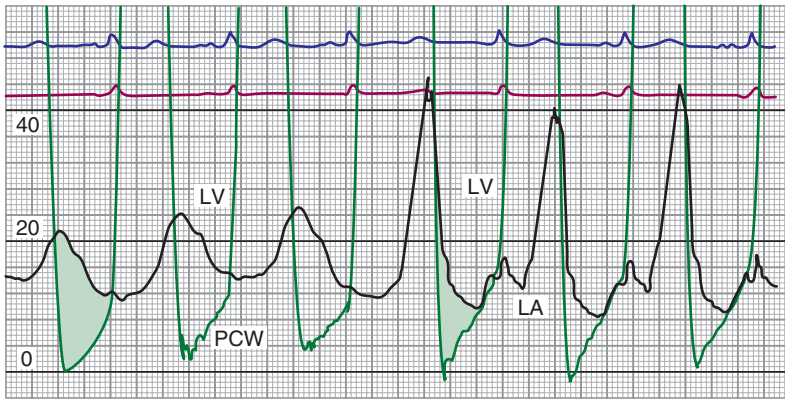


Fig. 5.6 Use of the pulmonary capillary wedge pressure can overestimate the transmitral gradient compared with the left-atrial pressure because there is a time delay with the pulmonary capillary wedge pressure and a dampening effect on the *v* wave, as shown here. *LA*, left atrial; *LV*, left ventricular; *PCW*, pulmonary capillary wedge. (From Syed Z, Salinger MH, Feldman T. Alterations in left atrial pressure and compliance during balloon mitral valvuloplasty. *Catheter Cardiovasc Interv*. 2004;61:571–579.)

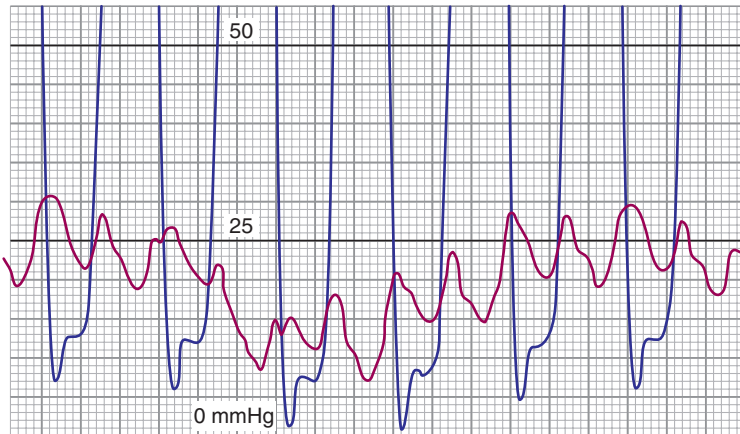


Fig. 5.7 Example of a transmitral gradient from severe mitral annular calcification in a patient with calcific aortic stenosis.

Box 5.2 Causes of Gradient Between Pulmonary Capillary Wedge and Left-Ventricular Diastolic Pressure

1. Mitral valve stenosis
2. Mitral annular calcification
3. Severe mitral regurgitation
4. Atrial myxoma (rare)
5. Cor triatriatum or pulmonary venoocclusive disease (very rare)
6. Hemodynamic artifacts
 - Improper zeroing, transducer balancing, or calibration
 - Pulmonary artery catheter not in true “wedge” position
 - Large *v* waves on pulmonary capillary wedge pressure

when invasive assessment is performed with transseptal catheterization, the transmitral gradient is found to be significantly less than the value measured by echocardiography (Fig. 5.8). The hemodynamics of mitral annular calcification is described in more detail later in this chapter.

A pressure gradient may be present in patients with severe MR (averaging about 6 mm Hg) because of the marked increase in flow across the valve, but it is observed only in early diastole.⁵

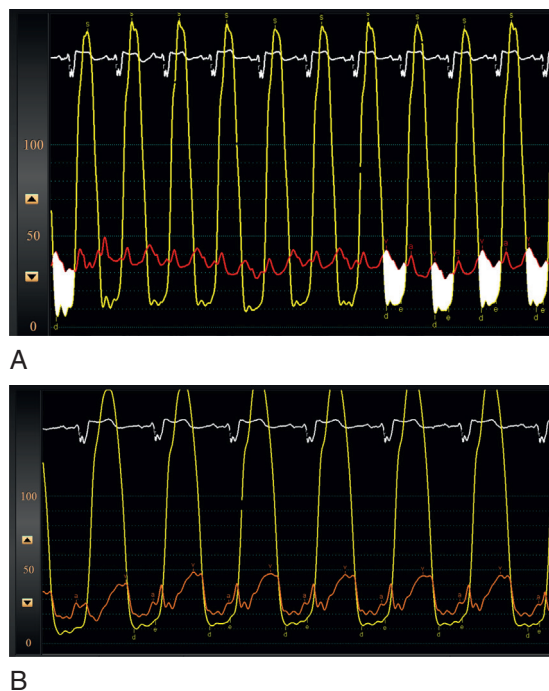


Fig. 5.8 This patient was a 77-year-old woman with heart failure syndrome, normal systolic function, and evidence of severe mitral annular calcification on echocardiography. The mitral valve gradient was estimated to be 20 mm Hg by echocardiography and the patient was thought to have severe mitral stenosis. On cardiac catheterization, there was severe pulmonary hypertension (pulmonary artery 75/28 mm Hg), and the simultaneous pulmonary capillary wedge and left-ventricular pressure suggested severe stenosis. (A) Because pulmonary hypertension was present, a transseptal catheterization was performed, and a much smaller gradient was observed. (B) This indicated mild stenosis but high left-atrial pressure from a large *v* wave due to severe diastolic dysfunction.

More important are several artifactual conditions resulting in an apparent transmitral gradient. Meticulous attention to detail is important when making these measurements, and pressure transducers should first be carefully leveled, calibrated, and zeroed. Because the pressures under consideration are relatively low, small errors in zeroing, improper catheter flushing, transducer level, or differences in frequency response between the two transducers may cause the false appearance of a transmitral gradient. Probably the most common artifact is due to the inability of the operator to achieve a true “wedge” position, particularly when severe pulmonary hypertension is present. In this scenario the pressure wave represents a hybrid between the true wedge pressure and the pulmonary artery systolic pressure, falsely causing or elevating the gradient (Fig. 5.9). Because of the time delay inherent to the generation of the PCWP waveform, a large *v* wave will appear within diastole and give the impression that there is an elevated gradient (Fig. 5.10). A transseptal catheterization with direct measurement of left-atrial pressure will demonstrate the absence of stenosis (Fig. 5.11).

CALCULATION OF MITRAL VALVE AREA USING THE GORLIN FORMULA

Richard Gorlin¹⁷ and his father derived the Gorlin equation in 1951 based on the physics of hydraulic systems. First, Gorlin chose the hydraulic formula for determining the area of a “rounded edge” orifice:

$$F = CcAV$$

where *F* = flow, *A* = orifice area, and *V* = the change in velocity of flow across the orifice. The value *Cc* represents the coefficient of orifice contraction to allow for the contraction of the stream as it passes through the orifice. This formula is then combined with the relationship:

$$V = Cv\sqrt{2gh}$$

where *g* = the gravitational constant (980 cm/s²) and *h* = the height of the column of fluid and can be substituted for the pressure gradient across the orifice. The value *Cv* represents the coefficient of velocity to account for the loss of

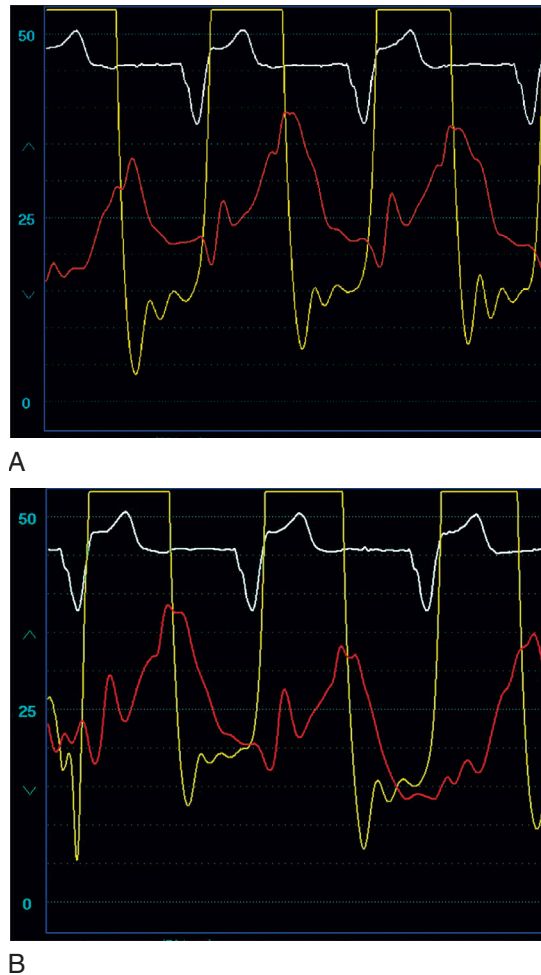


Fig. 5.9 A common source of error in the identification and quantification of a transmitral gradient when the pulmonary capillary wedge pressure is used instead of the left-atrial pressure. (A) A suitable wedge pressure and a significant v wave with an end-diastolic gradient of approximately 8 mm Hg suggest mitral stenosis. However, a blood sample drawn from the catheter in this position revealed an oxygen saturation of 75%. (B) The catheter was repositioned and wedge position was confirmed with an oxygen saturation of 95%; in this case, an end-diastolic pressure gradient is no longer present.

energy through friction and turbulence. The two formulas are then combined and simplified and a single empiric constant "C" is created to account for the various coefficients to produce the essence of the Gorlin formula, which states:

$$\text{Valve area} = \frac{\text{Value flow}}{C(44 \cdot 5) (\sqrt{\text{Pressure gradient}})}$$

Mitral valve flow is defined as the flow that occurs during diastole (diastolic filling period). Gorlin determined the empiric constant as 0.7 by collecting the hemodynamics from a single patient with mitral stenosis, measuring the actual valve area by autopsy after the patient died, and solving the formula for C. Because Gorlin had no method to measure LVEDP, he assumed a value of 5 mm Hg and determined the diastolic filling period from the brachial artery tracing as the beginning of the diastolic notch to the beginning of the upstroke of the next pressure pulse. Validation consisted of measurements obtained in 11 patients (6 autopsy and 5 surgical cases) with good correlation. Once LVEDP

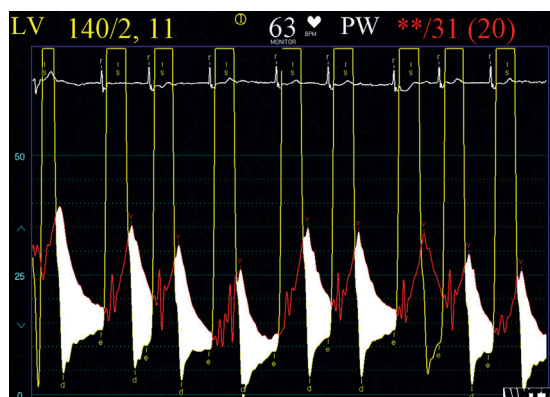


Fig. 5.10 This patient had very large v waves on the pulmonary capillary wedge pressure tracing and created the appearance of a mitral gradient when simultaneous left-ventricular and wedge pressures were recorded. LV, Left ventricular; PW, pulmonary capillary wedge pressure.

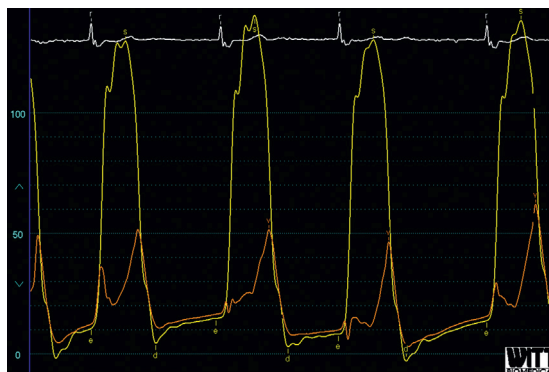


Fig. 5.11 A transseptal catheterization was performed in the patient shown in Fig. 5.10 and confirmed that there was no significant mitral stenosis. There is a large v present on the left-atrial waveform but no gradient across the mitral valve.

could be measured routinely, the Gorlin constant was corrected from 0.7 to 0.85, and the Gorlin formula evolved into its present form¹⁸:

$$\text{Mitral Valve area} = \frac{\text{Cardiac output}}{(\text{Diastolic filling period}) (\text{Heart rate}) (37.9) \sqrt{\text{Pressure gradient}}}$$

The pressure gradient represents the mean gradient and, in the current era, is typically measured with automated, computer-based hemodynamic systems. In normal sinus rhythm, five cardiac cycles are averaged. Because of the marked variation in the gradient with varying RR intervals, at least 10 cardiac cycles are required for patients with AF (Fig. 5.12A). Note the effect on the gradient following the compensatory pause after a premature ventricular contraction, as shown in Fig. 5.12B. The longer duration of diastole allows more time for the atrium to empty and reduces the end-diastolic gradient. As noted earlier, if the patient is in sinus rhythm and the PCWP is used instead of left-atrial pressure, the presence of a large v wave can falsely add to the gradient.

A much-simplified version of the Gorlin formula for calculating valve area has been proposed.¹⁹ The formula is easy to remember and eliminates the heart rate, diastolic filling period, and empiric constant:

$$\text{Valve area} = \frac{\text{Cardiac output}}{\sqrt{\text{Pressure gradient}}}$$

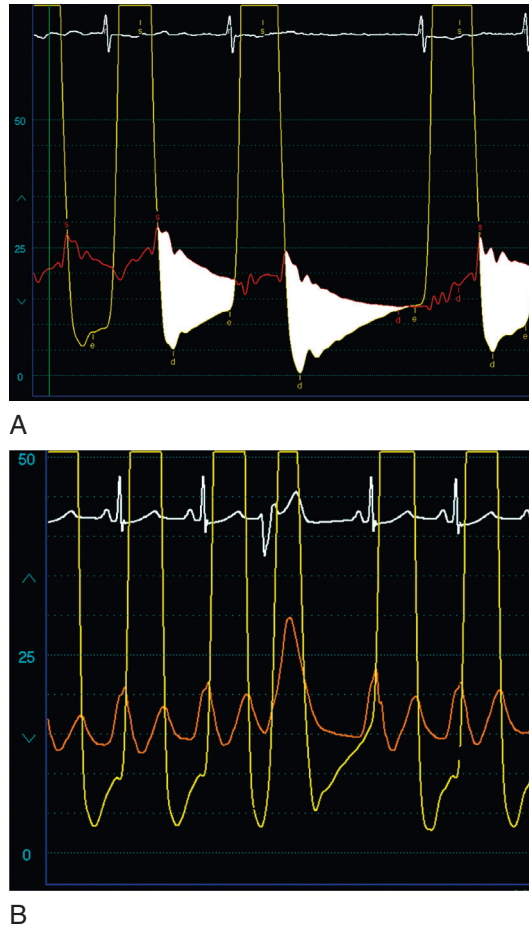


Fig. 5.12 The heart rate greatly impacts the transmitral gradient. (A) In atrial fibrillation varying RR intervals occur and the transmitral gradient is greatest when there is a short RR interval; longer RR intervals allow more time for the atrium to empty, and thus diastasis is achieved. (B) In patients with normal sinus rhythm the compensatory pause following a premature ventricular beat will prolong the RR interval and diminish the end-diastolic pressure gradient, also allowing diastasis.

When compared to the traditional Gorlin formula, the simplified formula may lead to significant disparity, especially if tachycardia is present (heart rate > 100 beats/min).²⁰ Therefore the simplified formula should be used with great caution.

The Gorlin formula for valve area calculation has several well-known criticisms and limitations. The formula is based on idealized relationships between flow across valves and orifice area and makes many assumptions and oversimplifications to create a tidy mathematical formula. The formula works best if normal sinus rhythm is present and is within the normal physiologic range of flows. Coexisting MR represents a significant limitation of this method as the Gorlin formula underestimates the true valve area because the cardiac output entered in the numerator is the forward flow and does not account for the regurgitant fraction included in the total transmitral diastolic flow.

In the event that a hemodynamic evaluation reveals only mild to moderate stenosis but a more severe valve lesion is suspected clinically, hemodynamics can be done under exercise conditions in the cardiac catheterization laboratory. While sometimes awkward to perform, even just a few minutes of upper arm exercise can cause increased heart rate and cardiac output and may allow a more clear assessment of the severity of stenosis.²¹ This discrepant scenario can be evaluated further noninvasively by assessing the gradients across the mitral valve and pulmonary pressures utilizing an echocardiographic stress test (preferably physical exercise or alternatively using dobutamine). This is also useful when evaluating females with mitral stenosis in the moderate to severe range and who are asymptomatic, particularly for individuals who are contemplating pregnancy, as pregnancy generally leads to an increase in cardiac output, heart rate, transmitral flow, and total blood volume, each of which can worsen transvalvular mitral gradients.

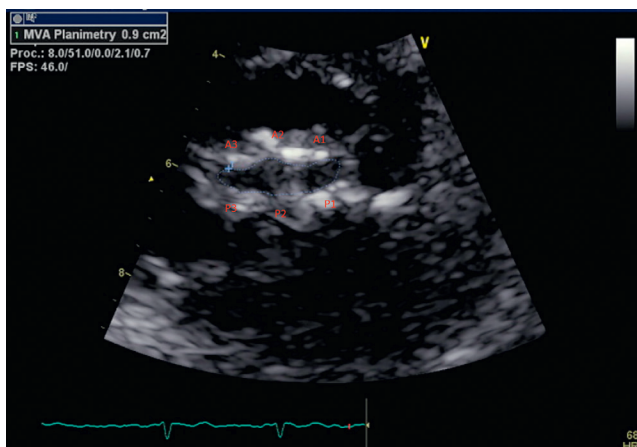


Fig 5.13 Planimetry of the mitral valve leaflet tips by 2D echo in the short-axis view. A1, A2, and A3 denote the three segments of the anterior leaflet, and P1, P2, and P3 denote the three segments of the posterior leaflet. In this case, the mitral area by planimetry is 0.9 cm^2 (dotted lines).

The severity of mitral valve stenosis is routinely evaluated noninvasively. Echocardiographic techniques include the use of two-dimensional (2D) echocardiography with planimetry of the mitral valve area and the use of Doppler echocardiographic techniques, including the pressure half-time and continuity equation methods. Although these techniques correlate reasonably well with the invasively determined techniques that rely on the Gorlin formula, considerable variability may exist between the methods in patients with symptomatic and significant mitral stenosis.^{22,23} This is particularly true in patients with lower transmitral gradients and higher cardiac outputs. Importantly clinicians should not fixate on the absolute value generated by these studies or rely solely on a single determination of the severity of stenosis. One series found that 12% of patients who underwent mitral valve surgery with severe symptoms and mitral stenosis had relatively small gradients ($<10 \text{ mm Hg}$) and only modest narrowing by calculated valve area (1.6 cm^2) yet improved dramatically with surgery, emphasizing the point that nothing substitutes for clinical judgment.²⁴

There are several echocardiographic findings characteristic of rheumatic mitral stenosis. First, there is commissural fusion with valvular thickening, particularly at the free edge of the leaflets. This is usually best seen in the parasternal short-axis view. Later thickening extends toward the base with further restriction of valve motion. Second, there is reduced leaflet mobility and restricted opening. The posterior leaflet is usually partially or completely immobile and the anterior leaflet shows a diastolic doming, producing the characteristic “hockey stick” configuration. This is usually best seen in the parasternal long-axis view. Third, there is involvement of the subvalvular apparatus with thickened, fused, and/or shortened chords. Finally, calcification is usually apparent of the leaflets and the subvalvular apparatus with more calcification seen in older patients with long-standing disease.

Direct planimetry of the mitral valve area is one method to evaluate mitral stenosis severity. 2D planimetry is performed in the parasternal short-axis view. It involves tracing the inner edge of the mitral valve orifice in mid-diastole. This measurement is made at the leaflet tips, ensuring the narrowest valve orifice. To select an adequate alignment of the valve orifice plane, a careful scanning from base to apex should be performed (Fig. 5.13). Zoomed images should be acquired, and an excessive gain setting should be avoided as it may lead to underestimation of the mitral valve area. 2D planimetry is a direct anatomic measurement. It is independent of load conditions, chamber stiffness, and concomitant valve disease and reliably correlates with the size of the anatomic orifice. Potential errors in this measurement include a more atrial level of tracing and thus missing the narrowest point of the valve, inadequate acoustic windows, and severe calcification or distortion of the leaflets. Among these limitations, inadequate alignment of the image plane is the main source of error. Due to these limitations, the severity of mitral stenosis should not be defined solely by this parameter, and an integrated approach is recommended. Three-dimensional echocardiography (3DE) is superior in evaluating mitral stenosis severity because the direct visualization of the valve orifice in multiple planes ensures that the planimetry takes place at the level of the mitral leaflet tips, where the valve orifice is narrowest. Limitations of 3DE include reduced spatial and temporal resolution compared to 2D echocardiography and suboptimal images in patients with arrhythmias.

The pressure half-time (PHT) method refers to the time required for the peak mitral gradient to drop by one-half. The more severe the stenosis, the slower the left ventricle fills and the longer the time required for the transmitral gradient to decrease (Fig. 5.14). This is represented invasively by the slow Y-descent on the LA or PCWP waveform (shown previously in Fig. 5.4). An advantage of this method is that it can be performed in cases where the acoustic window makes direct planimetry difficult. On the contrary, PHT is not dependent only on the mitral valve area but also on the

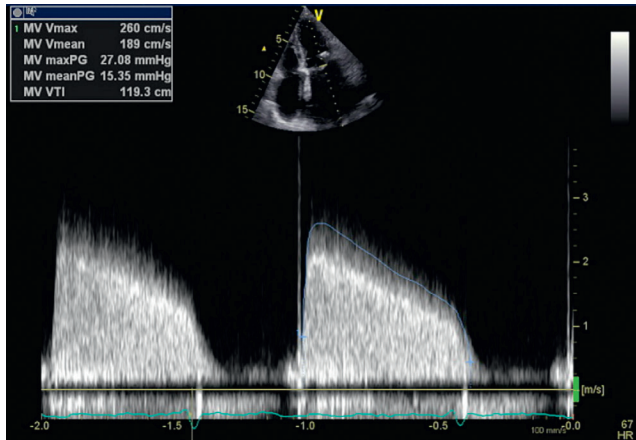


Fig 5.14 Example of a pressure half-time (PHT) determination by continuous wave (CW) Doppler. In this case, the maximum velocity is 260 cm/s, and the mean gradient is 14.35 mm Hg consistent with severe mitral stenosis.

left atrial and left ventricular hemodynamics and the peak mitral gradient. Thus in the presence of a restrictive filling pattern, low left atrium compliance, atrial septal defect, or significant aortic regurgitation, the PHT is shortened, and the mitral valve area is overestimated.

The mean gradient by Doppler echocardiography is the most important hemodynamic parameter to assess mitral stenosis severity. The peak gradient is more dependent on left atrial and left ventricular compliance and loading conditions and is therefore subject to more variability. Transmitral gradients are highly dependent on heart rate (the higher the rate, the higher the gradient). Changes in heart rate explain a substantial proportion of the variability. Therefore the heart rate at which the gradients are measured should always be reported. Similar to invasive assessment, in patients with AF, the mean gradient should be calculated as the average of at least five cycles with the least variation of R-R intervals and as close as possible to normal heart rate. The transmitral gradient is also influenced by flow, and in the presence of MR, may lead to an overestimation of the severity of mitral stenosis. The gradient in mitral stenosis should be differentiated from the gradient in advanced diastolic dysfunction where there is a gradient between the LA and LV since the LV filling depends on the elevated pressure LA; however, this gradient is present in the first half of diastole rather than throughout as in mitral stenosis as seen invasively (Fig. 5.2 and see Chapter 12).

HEMODYNAMIC FINDINGS OF MITRAL ANNULAR CALCIFICATION

In recent years, with the marked reduction in rheumatic heart disease, the epidemiology of mitral stenosis has evolved in developed countries, and there is now a greater prevalence of degenerative mitral stenosis related to mitral annulus calcification. This is generally seen in the elderly and patients with severe renal disease. It is regarded as an active and regulated molecular process of injury, lipid deposition, inflammation, and bone formation.²⁵ There are distinct echocardiographic features that differentiate rheumatic and degenerative mitral stenosis. As opposed to rheumatic stenosis, the main lesion is annular calcification, predominantly along the posterior aspect, resulting in a reduced annular area. Rarely, it extends anteriorly with severe forms involving the leaflets. Commissural fusion is not present and leaflet tip motion is relatively preserved. These forms are most often mild or moderate. The echocardiographic evaluation of degenerative mitral stenosis severity is difficult. As previously mentioned, rheumatic mitral stenosis severity can be assessed by planimetry, PHT, and the mean transmitral gradient. However, these parameters lack validation in degenerative mitral stenosis. Direct planimetry is challenging in these patients because of acoustic shadowing produced by calcium. Moreover, degenerative mitral stenosis is associated with advanced age, and diastolic dysfunction is common. PHT may lead to an overestimation of the mitral valve area in conditions of elevated left ventricle filling pressure or abnormal left atrial compliance.²⁶ The mean transmitral gradient is a useful parameter for evaluating degenerative mitral stenosis. In patients with mitral valve dysfunction related to annular calcification, the mean transmitral gradient is associated with increased mortality after adjustment for age, sex, and risk factors related to mitral annular calcification.^{27,28}

PROSTHETIC MITRAL VALVES

Evaluation of prosthetic mitral valves represents a unique challenge. Small gradients are present across most mechanical and biological mitral valve prostheses. The pressure gradient across normally functioning prosthetic valves depends on both the size of the prosthesis as well as its profile. The expected pressure gradient for many of the commonly used prosthetic valves is available and is based primarily on noninvasive techniques, using Doppler echocardiography.²⁹ In

general, by echocardiography, the normal velocity should be <1.9 m/s and the mean gradient <5 mm Hg; values exceeding these suggest stenosis of a mitral prosthesis.^{29,30}

In the cardiac catheterization laboratory, evaluation of a mitral prosthesis with suspected stenosis follows similar principles as described for a stenosed native valve. Usually transseptal catheterization is required for the highest accuracy. The degree of stenosis is typically described as the degree of pressure gradient. The use of the Gorlin formula to describe a prosthetic valve area is problematic given the earlier described limitations of the method. In the setting of the aortic valve, the Gorlin formula fails to accurately predict the prosthetic valve area³¹; similar inaccuracies in predicting the mitral valve area are likely. As stated earlier, the PCWP will tend to overestimate the transvalvular gradient, particularly if pulmonary hypertension is present; and many noted authorities recommend transseptal catheterization and direct measurement of left-atrial pressure in this setting, particularly if a decision regarding reoperation is entertained.

HEMODYNAMICS OF MITRAL BALLOON VALVULOPLASTY

Percutaneous mitral balloon valvuloplasty is considered the procedure of choice for the treatment of symptomatic, severe mitral stenosis, assuming anatomy is favorable and no contraindications exist, such as significant MR or atrial thrombus.³² The hemodynamic improvements are apparent immediately.^{33–37} Both the left-atrial pressure and the trans-mitral gradient fall and the cardiac output increases. The mitral valve orifice area increases on average from 1.0 cm² at baseline to a postprocedure valve area of 2.2 cm² and is similar to the results of open surgical commissurotomy.³⁴ An early improvement in pulmonary hypertension is augmented over time, although most patients with severe pulmonary hypertension will continue to have elevations of pulmonary artery pressure. If present, significant regression of tricuspid regurgitation occurs after successful mitral valvuloplasty in many cases.³⁶ Long-term results are favorable. The restenosis rate at 3 years is about 10% and is similar to open commissurotomy.³⁴ The 10-year survival, free of death, repeat procedure on the mitral valve, or class III or IV symptoms, is 56%.³⁵

An example of the typical hemodynamics seen in a patient with severe mitral stenosis undergoing successful mitral balloon valvuloplasty is shown in Fig. 5.15. The postvalvuloplasty left-atrial pressure dropped and the residual

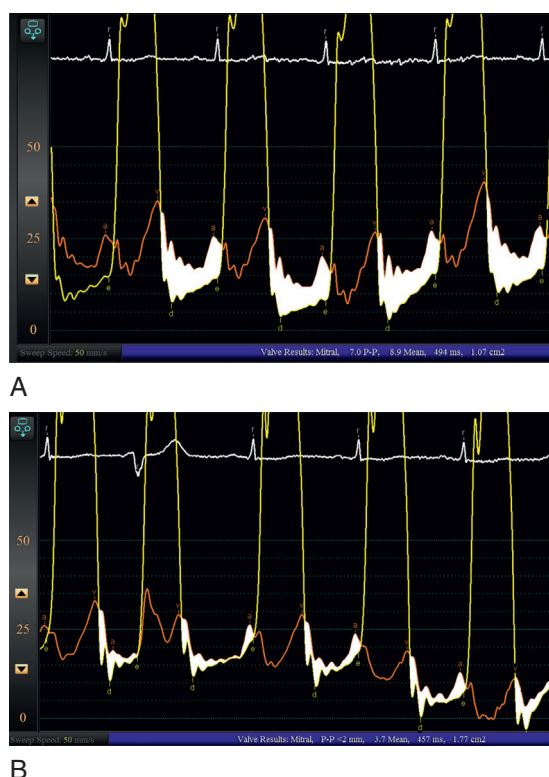


Fig. 5.15 (A) Simultaneous left-ventricular and left-atrial pressures confirm the presence of severe mitral stenosis in a 64-year-old woman with dyspnea on exertion and rheumatic deformity of the mitral valve. There is a mean gradient of 8.9 mm Hg and a valve area of 1.07 cm². (B) The results of mitral balloon valvuloplasty with a mean gradient of only 3.7 mm Hg and valve area of 1.8 cm². d, Diastole; e, end diastole.

transmitral gradient is minimal. Hemodynamic assessment postprocedure can also identify complications of the procedure. In the case shown in Fig. 5.16, balloon valvuloplasty was performed for severe mitral stenosis. The postvalvuloplasty hemodynamics show a rise in the left-atrial pressure with the appearance of a large and pathologic *v* wave consistent with the development of acute, severe MR, in this case due to tearing of the mitral leaflet during the procedure; this required urgent mitral valve replacement surgery.

MITRAL REGURGITATION

Regurgitation of the mitral valve occurs with much greater prevalence than stenosis. The term *primary mitral regurgitation* refers to regurgitation caused by an abnormality of the valve leaflets, annulus, or mitral apparatus. The term *secondary mitral regurgitation* is used to describe valvular regurgitation in the setting of reduced ventricular function, with left-ventricular dilation or prior myocardial infarction causing distortion or tethering of an otherwise normal mitral valve and apparatus leading to incomplete valve closure and regurgitation.³⁸ Box 5.3 lists the most common etiologies of MR with the most common cause in the current era due to myxomatous mitral valve disease. Although ischemia may cause infarction and subsequent avulsion of the papillary muscle and acute severe MR, the most likely mechanism of MR in the setting of ischemia is from secondary MR due to incomplete mitral leaflet closure from associated left-ventricular dysfunction and not from *papillary muscle dysfunction*.³⁹

The assessment of MR should start with three questions in mind: what is the mechanism; what is the severity; and what are the hemodynamic consequences of MR?

The mechanism MR is determined by a careful anatomic review using echocardiography. As stated earlier, primary MR indicates structural abnormalities of the leaflets themselves or the subvalvular apparatus (i.e., ruptured chordae),



Fig. 5.16 In this case mitral balloon valvuloplasty was performed in a 35-year-old woman with severe mitral stenosis due to rheumatic heart disease. The baseline hemodynamics are shown. (A) Simultaneous left-ventricular and left-atrial pressures confirm the presence of severe mitral stenosis. Following balloon valvuloplasty, the patient developed hypotension and became dyspneic. (B) On hemodynamics there is now a large and prominent *v* wave present on the left-atrial waveform (arrow) along with a drop in left-ventricular systolic pressure. In this case balloon valvuloplasty resulted in severe mitral regurgitation postmitral valvuloplasty.

Box 5.3 Causes of Mitral Valve Regurgitation**Primary Mitral Valve Regurgitation**

Myxomatous degeneration
 Spontaneous chordal rupture
 Papillary muscle rupture post-MI
 Rheumatic valve disease
 Endocarditis
 Collagen vascular diseases
 Hypertrophic obstructive cardiomyopathy (systolic anterior motion)
 Mitral annular calcification
 Prosthetic mitral valve dysfunction
 Prosthetic valve paravalvular leak

Secondary Mitral Valve Regurgitation

Mitral annular dilation due to poor LV function
 Tethering of the posterior leaflet from myocardial infarction
 Ischemic mitral regurgitation

MI, Myocardial infarction; LV, left ventricular.

Box 5.4 Grading the Severity of Mitral Valve Regurgitation

- 1+ Contrast clears with each beat, never filling the left atrium
- 2+ Contrast faintly fills the left atrium, but not as densely as the left ventricle, and does not clear with each beat
- 3+ Contrast completely opacifies the left atrium, ultimately, as densely as the left ventricle
- 4+ The left atrium becomes as densely opacified as the left ventricle, with the first beat becoming progressively denser with each beat and filling the pulmonary veins

while secondary MR is due to abnormal coaptation of normal leaflets most commonly caused by LV dysfunction, which in turn causes reduced closing force on the leaflets combined with systolic tethering of the leaflets into the LV, either by global LV dilation and prior infarction with focal wall motion abnormalities with preserved global LV function (i.e., infero-basal akinesis or dyskinesis). Secondary MR can also be caused by LV dyssynchrony due to bundle branch block or right ventricular (RV) pacing. In addition, abnormal leaflet coaptation and MR may be caused by mitral annular dilation due to left atrial dilation in chronic AF or restrictive cardiomyopathy. This is known as atrial functional MR. Finally, it is possible to have multiple mechanisms of MR with some features of primary and some features of secondary MR (mixed MR), especially in elderly patients with fibrocalcific changes in the valve.

The severity of MR is determined by a variety of methods; transthoracic or transesophageal echocardiography (TEE) are the main modalities but additional tools, including 3D echocardiography, cardiac computed tomography, and cardiac magnetic resonance imaging. In the cardiac catheterization laboratory, MR is quantified by contrast ventriculography using a semiquantitative scale (Box 5.4). This method is clinically useful but dependent on obtaining a high-quality ventriculogram. Emphasis on several important technical issues can improve the physician's ability to grade MR with this technique. Careful attention to proper catheter position will limit the degree of artifactual MR either from provocation of a ventricular arrhythmia or from entanglement of the pigtail catheter into the mitral apparatus, causing the operator to overestimate the true degree of regurgitation. Underestimation of the degree of regurgitation may occur if the catheter is placed too high in the left-ventricular outflow tract, underfilling the ventricle, or in the presence of chronic MR and/or chronic AF when there is marked enlargement of the left atrium. In such cases, contrast may not opacify the giant left atria, despite severe regurgitation, and may lead the operator to define a lower grade of regurgitation. The use of larger volumes of contrast than normally used (i.e., 50–60 mL) may remedy the problem. Finally, inadequate right-anterior oblique angulation may superimpose the left atrium over the descending aorta, making it difficult to accurately grade the degree of regurgitation.

Echo Assessment of Severity of Mitral Regurgitation and Hemodynamic Correlations. There are numerous methods using echocardiography to quantify the severity of MR.

1. **Color Doppler.** Doppler echocardiography is the most sensitive modality for the detection of MR. Among Doppler modalities, color Doppler is the initial screening method, localizing the origin of MR and its direction into the LA, and giving an overall gross assessment of its severity. However, color Doppler should not be recommended as the sole method for assessment of MR severity unless the degree of MR is obvious (i.e., trivial or massive amount of MR) and requires corroboration with other echocardiographic and Doppler methods.
2. **Regurgitant jet area.** Determination of the size of the regurgitant jet in the LA is a good, initial screening parameter for the severity of MR. However, note that the area of MR identified in the LA can mislead the severity of MR; this is because the color map is not a reflection of volume but of velocity.⁴⁰ Eccentric jets that impinge on the LA wall can

be small and thin and yet be severe, while central jet area may at times overestimate the severity of regurgitation because of the entrainment of blood cells around the jet.

3. **Pulsed wave Doppler** is also used to quantify MR. In patients with significant MR, the high flow across the MV results in a dominant early mitral inflow velocity (E), usually >1.2 m/s. The presence of an atrial (A) dominant mitral inflow virtually excludes significant MR since most of the flow occurs early in diastole. The prominent E-wave is seen invasively as a rapid Y-descent. However, a dominant E velocity is not specific for significant MR as it is elevated in heart failure with high filling pressures and mitral annular calcifications.
4. **Pulmonary venous assessment.** As regurgitation becomes more significant the systolic velocity of the pulmonary venous flow becomes progressively blunted, ultimately showing systolic flow reversal. Flow reversal in the pulmonary veins is specific but not sensitive to severe MR. Because regurgitant jets can be eccentric, they can selectively penetrate a particular vein according to the jet direction. Therefore it is essential to sample multiple pulmonary veins. A limitation of this parameter is that pulmonary vein systolic flow blunting can also be seen in AF or cases of elevated LA pressures.⁴¹
5. **Continuous wave Doppler (CWD).** The signal density by CWD is a qualitative measure of MR severity: a dense signal corresponds to a higher MR grade. A dense jet with a contour that is triangular and early peaking is the result of a high LA pressure with a prominent regurgitant pressure wave and usually signals significant MR. The triangular jet correlates invasively with the prominent V-wave. The MR jet peak velocity can also provide clues as to the severity of the hemodynamic state. In most cases, peak MR velocity is in the 5 to 6 m/s range because of the elevated LV-LA gradient. A lower MR peak velocity (and therefore a low LV-LA gradient) suggests either a low systemic blood pressure and/or a high LA pressure (Fig. 5.17).
6. **Vena contracta.** The vena contracta is the narrowest portion of the regurgitant flow that occurs at or immediately downstream of the regurgitant orifice. It is characterized by high-velocity laminar flow and is slightly smaller than the anatomic regurgitant orifice. The cross-sectional area of the vena contracta represents a measure of the effective regurgitant orifice area, a true parameter of lesion severity and is more accurate than vena contracta diameter (Fig. 5.18).

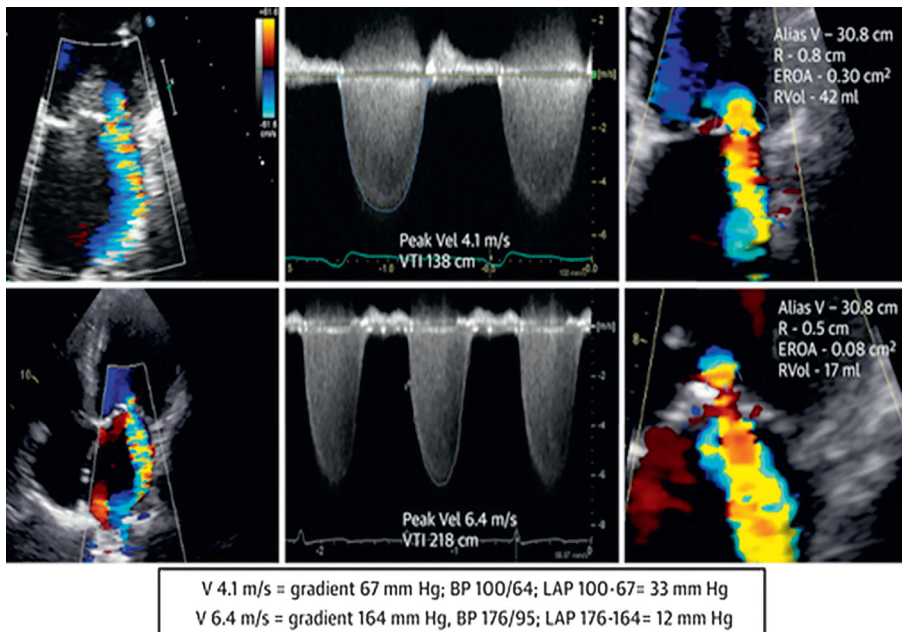


Fig. 5.17 Images from two patients (top and bottom) with functional mitral regurgitation (MR) and color Doppler jets that are quite similar in appearance (left panels). However, the continuous wave signal is 4.1 m/s in the patient on top and 6.4 m/s in the patient on bottom. This results in a calculated effective regurgitant orifice area and regurgitant volume in the moderately severe (3+) range for the top patient and mild (1+) for the bottom patient. Left-atrial pressure can be estimated from the systolic blood pressure minus the peak gradient calculated from the MR jet peak velocity (legend below bottom panels) and is elevated in the patient with 3+ MR and normal in the patient with 1+ MR. (From Grayburn PA, Thoma JD. Basic principles of the echocardiographic evaluation of mitral regurgitation. *JACC Cardiovasc Imaging*. 2021;14(4):843–853).

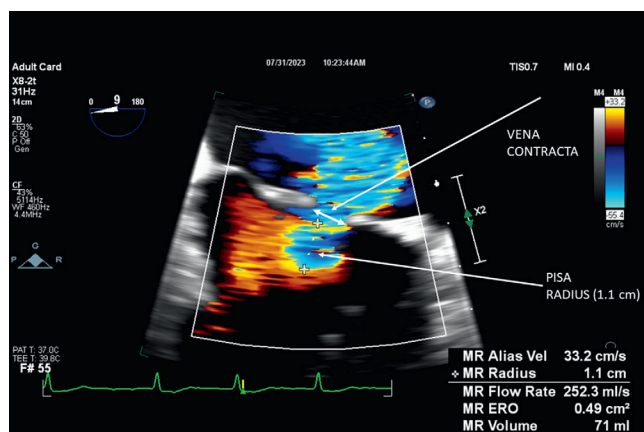


Fig. 5.18 Example of the vena contracta width (white arrow) within the mitral orifice and the PISA radius (between the two crosses and measuring 1.1 cm) in a patient with severe mitral regurgitation.

7. Proximal isovelocity surface area (PISA). PISA is a semiquantitative tool for the assessment of MR (Fig. 5.18).

It is based on the continuity equation, which is based on Gorlin formula where flow is the product of orifice area and velocity. Careful attention should be paid to assess whether the regurgitation covers the entire systole period, otherwise PISA, which is a semiquantitative method will overestimate the degree of MR severity.

Chamber size can provide some information about the severity of MR. For patients with chronic, primary MR, the volume overload will eventually lead to LA and LV chamber enlargement. In secondary MR, the relationship between MR severity and LA and LV dilation and dysfunction is difficult to ascertain because of the complicated interrelationship between these measures. In general, a normal LA size excludes severe chronic MR. On the other hand, an enlarged LA is not specific to severe MR as it can be the result of various disease states (e.g., hypertension, AF).

The hemodynamic consequences of MR depend on the severity, acuity of the onset of regurgitation, and compensatory responses. Severe acute MR (often due to spontaneous chordal rupture or avulsion of a papillary muscle head from acute infarction) typically occurs in the presence of a small, unadapted left atrium. The acute increase in volume abruptly raises left-atrial and pulmonary venous pressures, transmitting to the lungs and causing pulmonary edema. With a large proportion of the left-ventricular stroke volume deposited into the left atrium, forward flow is diminished and hypoperfusion and shock may ensue. Compensatory mechanisms are minimally effective and include tachycardia, in an effort to maintain cardiac output in the face of a diminished forward stroke volume, and peripheral vasoconstriction to maintain perfusion of vital organs. Acute, severe regurgitation represents a medical emergency with urgent surgical repair often indicated.

It is important to note that by echocardiography, the color Doppler jets seen in acute MR tend to be eccentric. This fact, along with the associated compensatory tachycardia and low driving pressure in an unstable patient, can make the diagnosis of severe MR difficult with transthoracic color Doppler. A hyperdynamic LV with low systemic output and low-velocity MR jet should raise suspicion about acute MR and prompt an assessment with TEE.⁴²

Some patients with acute MR may stabilize and enter a chronic, well-compensated phase. Alternatively, MR may begin as mild or moderate in extent and slowly progress, allowing for chronic compensation. Importantly MR tends to be progressive; thus moderate degrees of regurgitation often worsen because the additional stress on the mitral apparatus changes left-ventricular geometry, worsening regurgitation over time.

Chronic compensatory mechanisms allow for a prolonged asymptomatic state, despite severe MR. These mechanisms include the development of an increase in left-ventricular volume with the preservation of the ejection fraction to allow for a greater stroke volume to maintain forward flow, despite the regurgitant volume.⁴³ Eccentric hypertrophy balances wall stress from increased ventricular size; however, diastolic function remains normal to allow the increase in volume to occur without increasing left-ventricular diastolic pressures. Another important compensatory mechanism is the increase in left-atrial size to accommodate the regurgitant volume at lower filling pressures.

Over time, however, compensatory mechanisms fail. For poorly explained reasons, deterioration in contractile function occurs, resulting in myocardial dysfunction and a fall in ejection fraction. Although this phenomenon may be asymptomatic, it is often associated with symptoms of dyspnea from increased PCWP and/or fatigue from diminished cardiac output. Mitral valve repair or replacement is clearly indicated either on arrival of symptoms or with the onset of ventricular dysfunction. Ultimately, if untreated or allowed to advance, decompensation leads to left-heart failure and its consequences, including secondary pulmonary hypertension and right-heart failure.

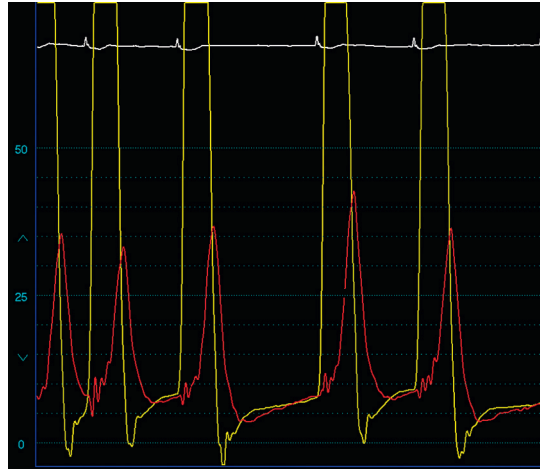


Fig. 5.19 Example of a large *v* wave on a pulmonary capillary wedge pressure in the complete absence of mitral regurgitation. Heart failure caused the large *v* wave in this patient.

HEMODYNAMICS OF MITRAL REGURGITATION

The classically taught hemodynamic abnormality of MR is the presence of a prominent *v* wave on the PCWP or left-atrial pressure tracing. However, prominent *v* waves are often absent, despite severe MR, and are frequently observed in conditions other than MR.

Recall that a *v* wave is a normal physiologic event due to atrial filling at the end of ventricular systole when the mitral valve is closed. An abnormal or prominent *v* wave has been variably defined as a peak *v* wave in excess of 40 mm Hg, a difference between the peak *v* wave and mean PCWP >10 mm Hg, or the ratio of the peak *v* wave to mean PCWP >2.⁴⁴ Some have proposed that a *v* wave height three times the mean PCWP is virtually diagnostic of severe, acute MR.⁴⁵

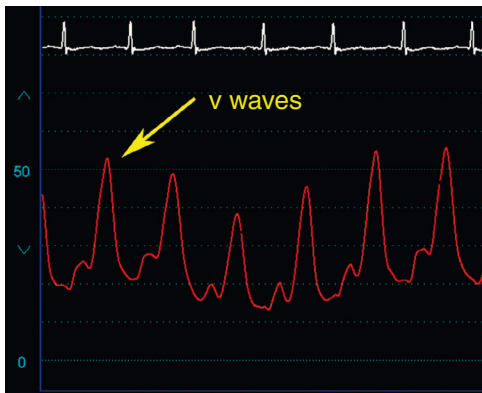
It has been long recognized that the presence or absence of an abnormal *v* wave fails to correlate with either the presence or severity of MR, which is particularly true in the setting of chronic MR. As far back as 1963, Braunwald⁴⁶ observed that severe, chronic MR could exist with normal left-atrial pressure and a normal-sized *v* wave. Several investigators have reported that about one-third of patients with prominent *v* waves had no MR.^{47,48} In one study of over 900 patients who underwent both ventriculography and right-heart catheterization, using a large lumen catheter and performance of an oximetrically confirmed wedge pressure measurement, the presence of a prominent *v* wave was insensitive and had a poor positive predictive value for the presence of moderate or severe MR.⁴⁴ Interestingly, these investigators found that the *absence* of a prominent *v* wave was 94% specific and had a 93% negative predictive value for the *absence* of severe MR. This study consisted mostly of patients with chronic MR; the correlation might differ with acute MR, in which the *v* wave is more likely to be prominent.

The poor correlation between the presence or extent of MR and the height of the *v* wave likely relates to the numerous physiologic factors involved in determining the *v* wave. These include (1) the rate and volume of blood that enters the left atrium during ventricular systole, (2) the volume and pressure of blood that exists within the left atrium, (3) systemic afterload that influences atrial emptying, (4) left-ventricular contractile force that affects both left-ventricular end-diastolic volume and pressure, and (5) left-atrial compliance. Accordingly, abnormal *v* waves are seen in MR and ventricular septal defects because of the increased volume that enters the left atrium. Similarly, large *v* waves are a prominent feature of both mitral stenosis and congestive heart failure (Figs. 5.3 and 5.19) because the left-atrial volume and pressure are high and a small additional increase in volume results in a greater increase in pressure, thereby generating a more prominent *v* wave. Tachycardia may result in *v* waves due to the shorter diastolic emptying period. Perhaps the most important variable, however, is left-atrial compliance. The small, noncompliant atria of most patients with acute, severe MR explains why prominent *v* waves are more commonly seen in these patients as compared with those with chronic MR, in which the left atrium is enlarged and more compliant. Conditions other than MR associated with diminished compliance of the left atrium are also associated with prominent *v* waves, including the postoperative state, rheumatic heart disease, and ischemia.

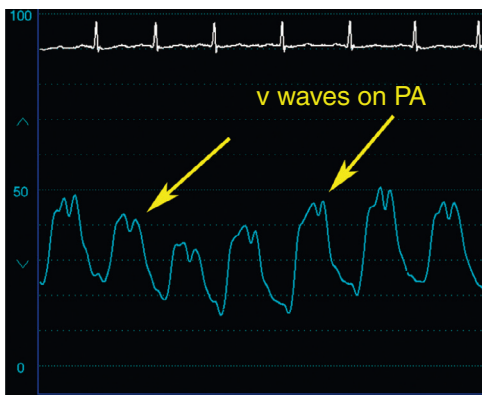
Patients with acute MR exhibit markedly abnormal hemodynamics. Hypotension, tachycardia, and a low cardiac output are often present. A marked elevation of the PCWP occurs, often with a very prominent *v* wave (Fig. 5.20). In fact, because the left atrium is typically small and noncompliant, the *v* wave may reach giant proportions and, in some dramatic cases, may even be apparent on the pulmonary artery waveform (Fig. 5.21). The transmission of this pressure



Fig. 5.20 Simultaneous left-ventricular and pulmonary capillary wedge pressure tracings obtained in a patient in cardiogenic shock from acute severe mitral regurgitation due to papillary muscle rupture in the setting of an acute, inferior wall myocardial infarction. A prominent v wave is present on the pulmonary capillary wedge pressure tracing and hypotension. A difference of only about 30 mm Hg exists between the peak of the v wave and the peak left-ventricular systolic pressure.



A



B

Fig. 5.21 Prominent v waves may be apparent on the pulmonary artery waveform. (A) In this case of severe acute mitral regurgitation, the pulmonary capillary wedge pressure demonstrated v waves that exceed 50 mm Hg in height. (B) Waves transmitted to the pulmonary artery systolic pressure waveform (arrows). PA, Pulmonary artery.

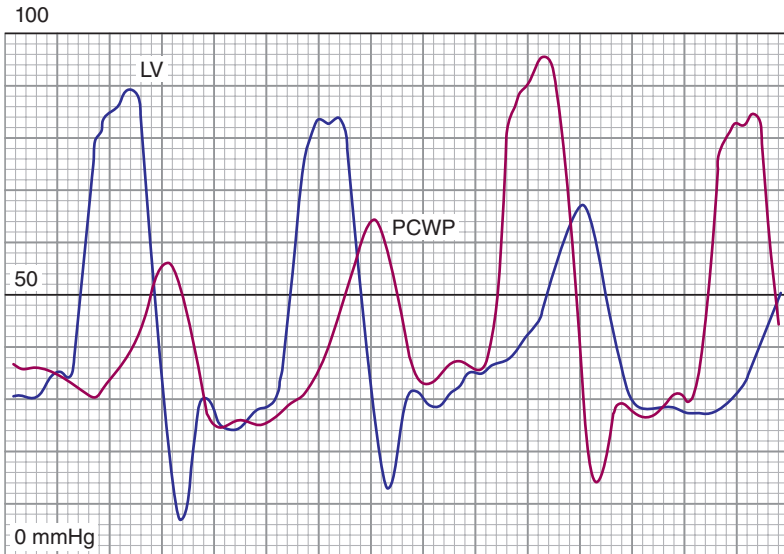


Fig. 5.22 Severe acute mitral regurgitation with cardiogenic shock, large *v* waves, and minimal pressure difference between the height of the *v* wave and the peak left-ventricular systolic pressure. In this case, no systolic murmur was appreciated. *LV*, Left-ventricular; *PCWP*, pulmonary capillary wedge pressure.

wave from the pulmonary veins to the pulmonary artery also explains the occasional phenomenon of a false elevation in the pulmonary artery saturation in some cases of severe MR.

In acute MR, the systolic murmur of MR may be absent or diminished, which is represented hemodynamically by the observation of a minimal gradient between peak left-ventricular systolic pressure and the height of the *v* wave on the PCWP tracing (Fig. 5.22). Another interesting finding can be observed in acute MR due to the constraining effect of the intact, normal pericardium in the presence of acute volume overload from acute MR, resulting in hemodynamic abnormalities similar to constrictive pericarditis, with elevated and equalized right-ventricular and left-ventricular diastolic pressures.⁴⁹

The hemodynamics seen in patients with chronic MR may, in fact, be entirely normal or only mildly abnormal at rest if ventricular function remains normal and they are well compensated (Fig. 5.23). As mentioned earlier, the height of the *v* wave is an unreliable indicator of the severity of MR, with one-third of patients with chronic MR demonstrating trivial *v* waves, despite severe MR⁴⁷ (Fig. 5.24). A small diastolic pressure gradient may be observed across the mitral valve; however, unlike in mitral stenosis, the gradient is present during early diastole only. In addition, the slope of the *y* descent in MR is steep rather than delayed, as seen in mitral stenosis.

Importantly, it should be noted that hemodynamic measurements in the cardiac catheterization laboratory are obtained under resting (and often sedated) conditions. Many patients with chronic MR experience symptoms of exertion. Hemodynamics obtained during dynamic exercise during cardiac catheterization may be more revealing. A dramatic rise in the wedge pressure or pulmonary artery systolic pressure during exercise may be revealing. In one study, about 20% of patients with normal resting PCWP developed *v* waves greater than 50 mm Hg with exercise.⁵⁰

With the onset of symptoms and the development of decompensation, derangements in hemodynamics become prominent and include the development of elevated left-atrial and pulmonary artery pressures. Again, a *v* wave may or may not be present. Over time and similar to other conditions that cause chronic elevations in the pulmonary venous pressure, secondary pulmonary hypertension with high pulmonary vascular resistance and subsequent right-sided heart failure may develop. In this setting secondary tricuspid regurgitation may arise from right-ventricular enlargement and annular dilation.

PERCUTANEOUS MITRAL VALVE REPAIR FOR MITRAL REGURGITATION

Traditionally severe MR was treatable only with open surgical techniques. Many patients could not be treated surgically because of significant comorbid conditions elevating their risk. For this reason, percutaneous techniques to treat the mitral valve are of intense interest and under vigorous development. Currently, “edge-to-edge repair” using either the MitraClip (Abbott Laboratories, Abbott Park, Illinois, USA) or the PASCAL device (Edwards Lifesciences, Irving, CA) is approved for patients with both primary and secondary (functional) MR. Details of this procedure have been described elsewhere.⁵¹ To summarize, the procedure is based on the Alfieri stitch, an open surgical technique to reduce

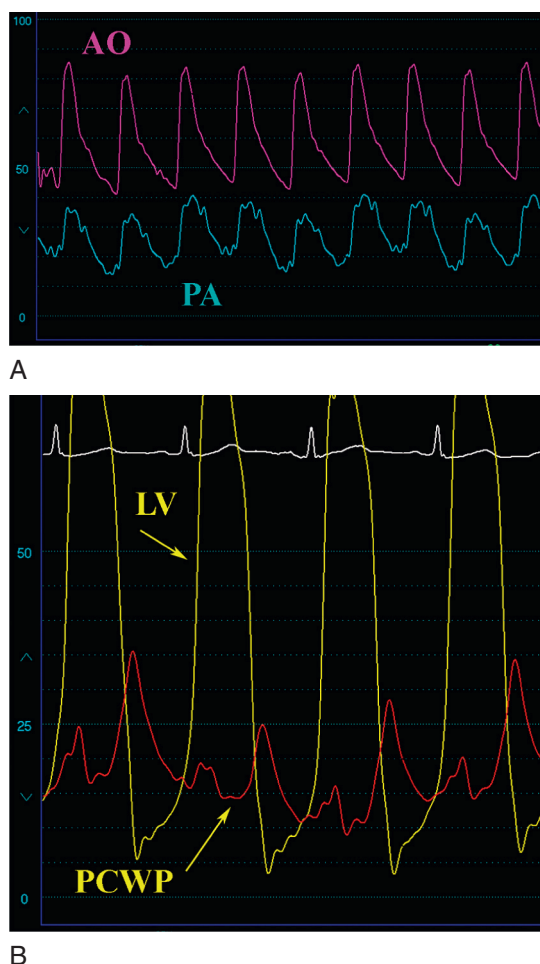


Fig. 5.23 Example of hemodynamics obtained in a patient with severe, chronic mitral regurgitation. (A) At rest, elevation of the pulmonary artery systolic pressures is moderate. The mean wedge pressure is normal, averaging 15 mm Hg. (B) The *v* wave reaches about 30 mm Hg. This patient has symptoms with exertion when the pulmonary artery and pulmonary capillary wedge pressure likely exceed these resting values. AO, Aortic; LV, left ventricular; PA, pulmonary artery; PCWP, pulmonary capillary wedge pressure.

degenerative MR by suturing together the tips of the mitral leaflets at a midpoint along the line of the mitral orifice and creating a double orifice. Transcatheter edge-to-edge repair (TEER) accomplishes this percutaneously in the cardiac catheterization laboratory via a venous approach across the atrial septum. The procedure is guided by TEE and fluoroscopy using proprietary clip delivery systems. These catheter-based procedures are safe and effective, with a durable effect.^{52–55} Although surgical methods are better at eliminating MR, the mitral clip is able to reduce the degree of MR and improve symptoms and quality of life in patients with severe MR, including those at prohibitive risk for surgery.^{54,56} Importantly, patients with functional MR treated with TEER have lower mortality and fewer admissions for heart failure as compared to those treated with medical therapy alone.⁵⁵

After the TEER procedure immediate improvements in hemodynamics are seen.⁵⁷ The most significant effects include an increase in cardiac output and forward stroke volume, with a corresponding reduction in systemic vascular resistance. Both LVEDP and volumes decrease. Interestingly although the *v* wave often decreases, there is no change in the mean PCWP, pulmonary artery systolic pressure, or pulmonary vascular resistance. This may be due to the fact that the TEER procedure also causes a small degree of stenosis in the creation of the double orifice. Alternatively, the true effect is not appreciated since the hemodynamics are typically collected during a resting state under general anesthesia. Following TEER, there may be a significant gradient across the mitral orifice. In fact, one study found that

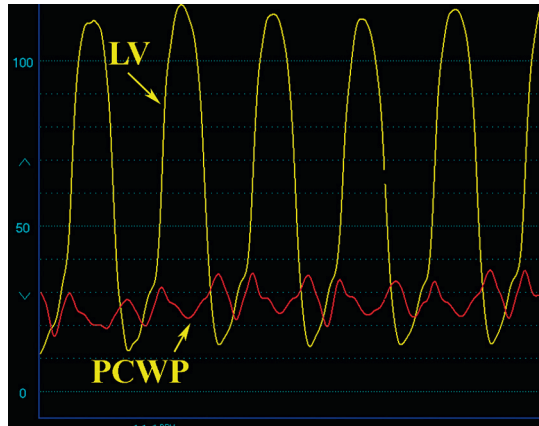


Fig. 5.24 Severe, chronic mitral regurgitation that demonstrates a normal-sized *v* wave on the pulmonary capillary wedge tracing. LV, Left ventricular; PCWP, pulmonary capillary wedge pressure.

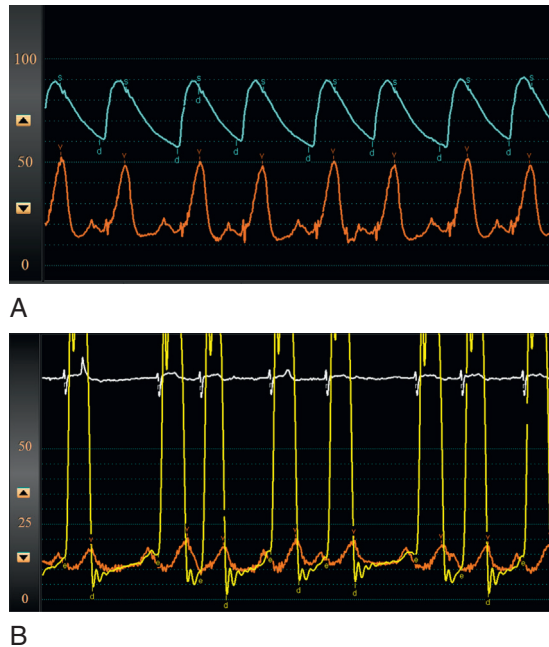


Fig. 5.25 These tracings were obtained in an elderly woman with severe mitral regurgitation due to prolapse, undergoing a percutaneous MitraClip procedure. (A) The baseline hemodynamics reveal marked elevation of the pulmonary artery pressure (70/29 mm Hg), and a large *v* wave of up to 50 mm Hg was present on the left-atrial waveform. (B) After successfully positioning the mitral clip, there was only mild mitral regurgitation present and the left-atrial pressure fell dramatically to a mean of only 12 mm Hg. Correspondingly, the pulmonary artery pressure fell to 48/23 mm Hg. *d*, Diastole; *e*, end diastole; *s*, systole.

37% of patients with degenerative (or primary) MR and 22% of patients with functional MR undergoing TEER had a significant gradient across the mitral valve (>5 mm Hg).⁵⁸ In this study, patients with degenerative MR and a >5 mm Hg transmitral gradient following TEER did not improve functional capacity. However, other studies have shown no relation to outcomes, and the significance of these residual gradients remains a source of debate.⁵⁹

An example of the hemodynamics obtained during a MitraClip procedure is shown in Fig. 5.25.

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